Republic of the Philippines

Western Mindanao State University

College of Computing Studies

Department of Information Technology

In Partial Fulfillment of the Requirements for the Degree Bachelor of Science in Information Technology

Website:

EDU SOLVE

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Table of Contents

CHAPTER 1 4
INTRODUCTION 4
1.1 Project Background 5
1.2 Purpose and Objectives 6
1.3 Significance 7
1.4 Target Audience 7
CHAPTER 2 9
OVERVIEW 9
2.1 Challenges Addressed 9
2.2 Solution Overview 9
CHAPTER 3 10
SYSTEM FEATURES 10
3.1. Interactive Problem-Solving Exercises 10
3.2 Engaging Lectures 15
3.3 Progress Tracking 18
3.4. Unlockable Lectures 20
3.6 Account Management 22
CHAPTER 4 24
SYSTEM ARCHITECTURES 24
4.1 System Workflow 24
4.2 Front-End Architecture 24
4.3 Back-End Architecture 30
4.4 Database Design 31
4.4.1 Key Components of the Database 31
4.4.2 Relationships Between Tables 31
4.4.3 Entity-Relationship Diagram (ERD) 32
4.4.4 Schema (Table Structure) 34
4.4. 5 Visual Representation in the ERD: 36
4.4.6 Query Optimization 37
4.4. 7 Relationships Between Tables: 37
CHAPTER 5 38
METHODOLOGY 38
5.1. Development Approach 38

5.2. Phases of Development 38
5.3. Tools and Technologies Used 39
5.4. Iterative Process 39
CHAPTER 6 40
DEPLOYMENT STEPS 40
6.1 User Guide 40
6.2 Web Hosting Setup 40
6.3 Platform Testing and Quality Assurance 41
6.4 Launch 41
6.5 Ongoing Support and Maintenance 41
CHAPTER 7 43
CONCLUSION 43

CHAPTER 1

INTRODUCTION

1

he Philippine education system stands at a critical juncture, grappling with persistent challenges in improving students' proficiency in essential academic subjects such as Mathematics, Science, and Reading. These challenges were starkly highlighted in the 2022 PISA (Programme for International Student Assessment) results, where the Philippines ranked among the lowest-performing countries globally. Such outcomes underscore systemic deficiencies and the urgent need for innovative solutions that transcend traditional teaching methods to bridge deep-rooted learning gaps.

EDU SOLVE rises as a transformative response to this pressing educational crisis. Tailored specifically for Grade 10 students, this web-based platform leverages the power of technology to create an engaging, personalized, and adaptive learning environment. By offering interactive problem-solving exercises, real-time progress tracking, and immersive multimedia lectures, EDU SOLVE empowers learners to strengthen their academic skills and rebuild their confidence in learning.

More than just a tool for academic improvement, EDU SOLVE embodies a broader mission to elevate the overall quality of education in the Philippines. Through the integration of modern pedagogical techniques and data-driven insights, it seeks to align with national educational reforms while addressing the unique needs of Filipino students. EDU SOLVE aims to create a learner-centered environment where students feel motivated to learn, understand foundational concepts before advancing, and build long-term academic success. Ultimately, EDU SOLVE envisions a future where Filipino learners thrive not only within their local academic environments but also on the global stage—becoming a generation equipped to compete, innovate, and succeed in an increasingly interconnected world.

1.1 Project Background

The education sector in the Philippines faces significant challenges, particularly in enhancing students' proficiency in fundamental academic subjects such as Mathematics, Science, and Reading. Despite ongoing efforts to improve the quality of education, the results of the 2022 Programme for International Student Assessment (PISA) revealed that the Philippines ranked among the lowest-performing countries globally. This highlighted not only the need for improved teaching methods but also a more engaging and effective approach to learning.

In response to these challenges, EDU SOLVE was conceptualized as an innovative educational platform aimed at addressing the academic deficiencies faced by Grade 10 students in the country. The platform is designed to bridge the learning gaps by offering interactive problem-solving exercises, engaging multimedia lectures, and real-time progress tracking tools. EDU SOLVE strives to make learning more accessible, engaging, and tailored to the needs of students, providing them with the tools necessary to improve their academic skills and boost their confidence in their educational journey.

The platform's features, including Unlockable Lectures, Progress Tracking, and Interactive Exercises, are structured to build knowledge progressively. As students advance through the lessons, they unlock additional content based on their progress, ensuring they have mastered foundational concepts before moving on to more advanced topics. This approach not only ensures a personalized learning experience but also promotes self-paced learning, enabling students to take ownership of their academic growth. The features integrate modern educational strategies such as adaptive learning and gamification, creating an engaging environment that encourages students to actively participate in their academic journey.

EDU SOLVE aims to be more than just a learning tool; it is part of a larger vision to enhance the quality of education in the Philippines, aligning with national educational reforms and helping students become globally competitive. By integrating modern educational techniques and leveraging the power of technology, EDU SOLVE supports the transformation of the Filipino education system, empowering students to succeed academically and excel in a rapidly evolving world.

1.2 Purpose and Objectives

Purpose

The purpose of EDU SOLVE is to support the academic growth of Grade 10 students in the Philippines, addressing the challenges highlighted in international assessments like PISA. This platform aims to provide an innovative, interactive, and adaptive learning environment that empowers students to improve their academic skills. Through engaging educational materials and comprehensive progress-tracking features, EDU SOLVE strives to equip learners with the tools they need to succeed in their studies and develop critical competencies. It focuses on fostering long-term academic achievement and preparing students for both national and international assessments by offering personalized learning pathways.

Objectives

- To improve the proficiency of Grade 10 students in core academic subjects such as Mathematics, Science, and Reading by offering interactive, personalized learning experiences.
- To provide students with a progressive learning path where they can track their academic progress and unlock additional learning content as they advance.
- To create an engaging and supportive environment that boosts students' confidence in learning and enables them to take ownership of their academic journey.
- To foster self-paced learning and encourage students to complete foundational tasks before advancing to more complex material, ensuring they build strong academic foundations.
- To help elevate the Philippines' education system by providing a modern, data-driven tool that aligns with national educational reforms and supports student success on a global scale.

1.3 Significance

The significance of EDU SOLVE extends far beyond being a mere educational platform; it is a catalyst for transforming the landscape of education in the Philippines. By addressing the fundamental academic challenges faced by students, particularly in core subjects like Mathematics, Science, and Reading, EDU SOLVE provides students with the skills and knowledge necessary to thrive academically. Through its interactive tools, personalized content, and measurable progress tracking, EDU SOLVE supports students in taking control of their learning journey.

This initiative is particularly critical in light of global education assessments such as PISA, where Filipino students have shown room for improvement. EDU SOLVE helps bridge this gap by offering an engaging and user-friendly learning experience that motivates students to succeed. By focusing on the core subjects that are crucial to a student's academic development, the platform allows learners to build confidence, strengthen their knowledge, and ultimately perform better in both local and international assessments.

Furthermore, EDU SOLVE contributes to the broader goal of improving the Philippines' standing in global education rankings. By providing students with modern learning tools and strategies, EDU SOLVE helps raise the bar for education in the country, ensuring that Filipino students are better equipped to face future academic and professional challenges. In doing so, it not only supports individual success but also contributes to the overall growth of the nation's educational system, preparing Filipino students to compete on the global stage.

1.4 Target Audience

The primary target audience for EDU SOLVE includes Grade 10 students in the Philippines who are struggling in key academic areas such as Mathematics, Science, and Reading. The platform is specifically designed to help these students improve their proficiency in these subjects, providing a personalized, engaging, and adaptive learning experience that helps them progress at their own pace.

Stakeholders include:

- **CHED (Commission on Higher Education):** Implementing EDU SOLVE as part of their efforts to improve educational standards across the country.
- **WMSU (Western Mindanao State University) Faculties:** Using EDU SOLVE as a supplementary tool in the classroom to enhance learning and help students improve their academic skills.

- **Educators and Parents:** Teachers and parents who are invested in supporting students' academic development and ensuring that they have access to the resources needed to succeed.

Additionally, EDU SOLVE can serve students from different learning backgrounds, including those who benefit from a more structured, progressive learning approach. It is designed to cater to the needs of a wide range of learners, helping them take ownership of their education and giving them the tools to succeed academically.

CHAPTER 2

OVERVIEW

2.1 Challenges Addressed

EDU SOLVE tackles critical challenges within the Philippine education system, including:

- The country's low ranking in the PISA 2022 assessment.
- Persistent difficulties faced by students in core subjects such as Mathematics, Science, and Reading.
- The urgent need for innovative tools to bridge learning gaps and address systemic deficiencies effectively.

2.2 Solution Overview

EDU SOLVE is an innovative web-based platform designed to enhance the learning experience for Grade 10 students. By integrating modern teaching methodologies with advanced technology, EDU SOLVE delivers an engaging suite of tools and resources tailored to address students' unique academic needs.

The platform features:

- **Interactive problem-solving exercises** to develop critical thinking skills.
- **Real-time progress tracking** to monitor student growth and provide actionable feedback.
- **Unlockable content** to motivate students to achieve milestones and sustain engagement.

This structured, student-centered approach ensures learners progress at their own pace while mastering foundational concepts before advancing. EDU SOLVE’s focus on fostering engagement and promoting self-paced learning empowers students to take ownership of their academic growth.

By addressing the root causes of learning challenges and equipping students with essential skills, EDU SOLVE sets a new benchmark for educational platforms in the Philippines, fostering academic excellence and future-ready learners.

CHAPTER 3

SYSTEM FEATURES

3.1. Interactive Problem-Solving Exercises

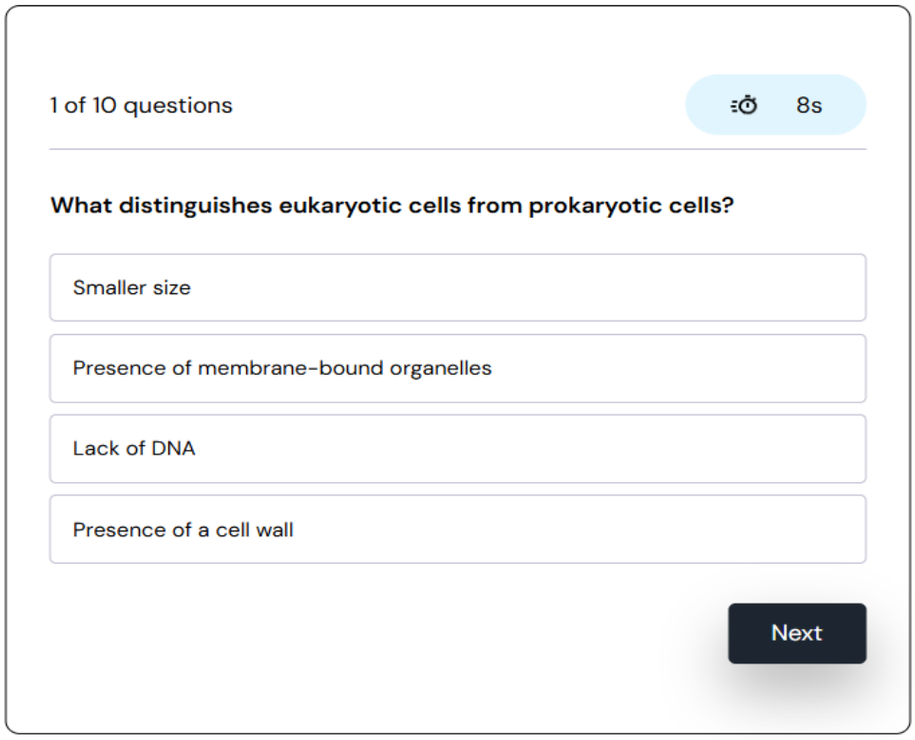
Our interactive problem-solving exercises are designed to enhance learning through active engagement, offering a hands-on approach to mastering core subjects. These exercises are tailored to improve critical thinking and problem-solving skills by presenting real-world scenarios and guiding users through step-by-step solution processes.

In addition to the exercises, we offer **Practice Tests** and **Final Tests** to assess progress and mastery of the material. The practice tests allow students to test their knowledge in a low-pressure environment, while the final tests evaluate overall understanding and readiness to move forward.

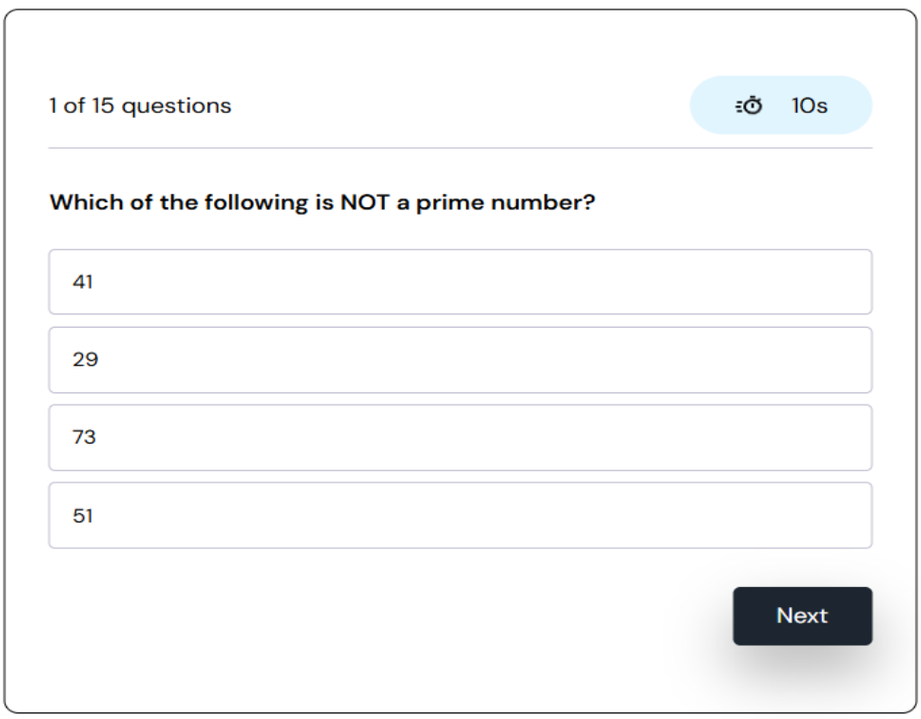
Key Features:

- **Practice Tests:**
  These tests provide users with an opportunity to apply what they have learned in a simulated exam format. By reinforcing knowledge through hands-on exercises, the practice tests help boost students' confidence and deepen their understanding of foundational concepts.

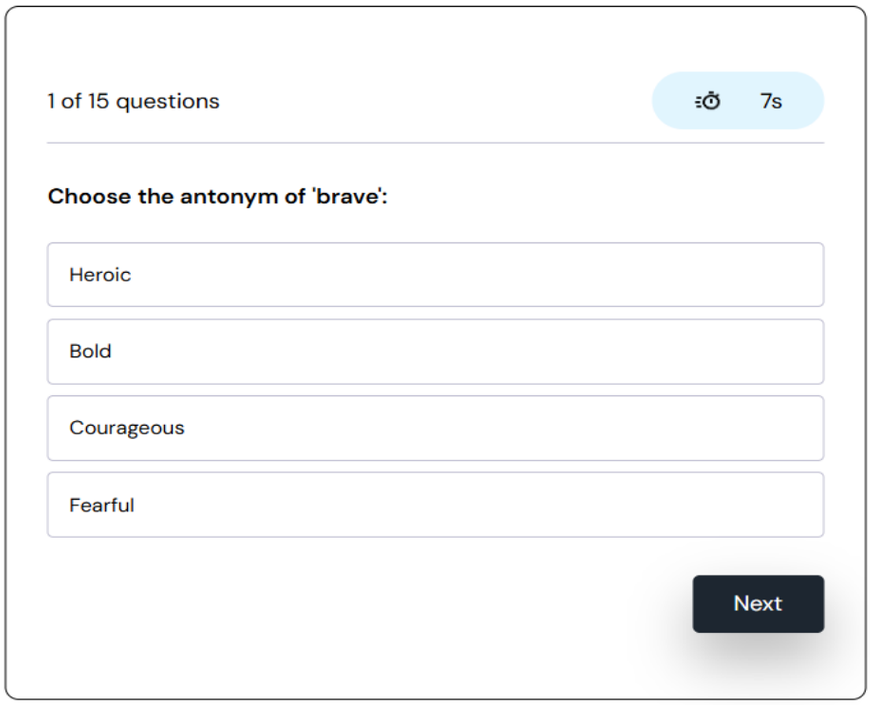
- ***Visual Representation:***
  Please see Figures 3.1.1a, 3.1.1b, and 3.1.1c for subject-specific views of the Practice Tests.
  These images illustrate the distinct layouts for Mathematics, Science, and Reading modules, showcasing how the system adapts problem-solving exercises to the unique requirements of each subject.



*Please see **Figure 3.1.1:** Overview of Interactive Problem-Solving Exercises (Practice Test) across Science, Mathematics, and Reading modules.*

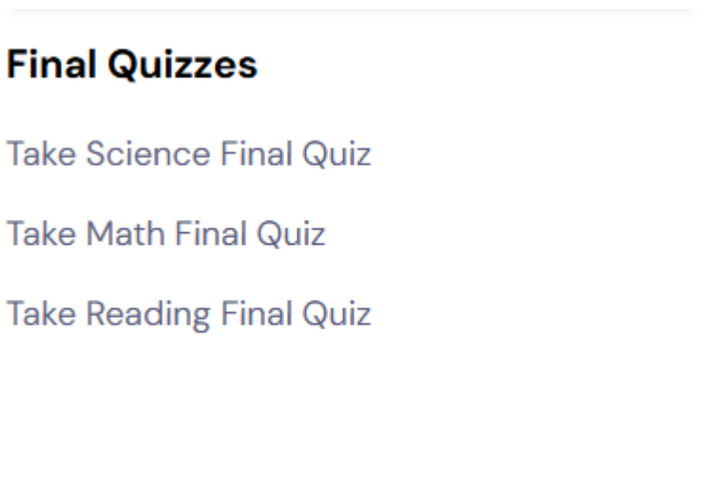


*Please see **Figure 3.1.1:** Overview of Interactive Problem-Solving Exercises (Practice Test) across Science, Mathematics, and Reading modules.*

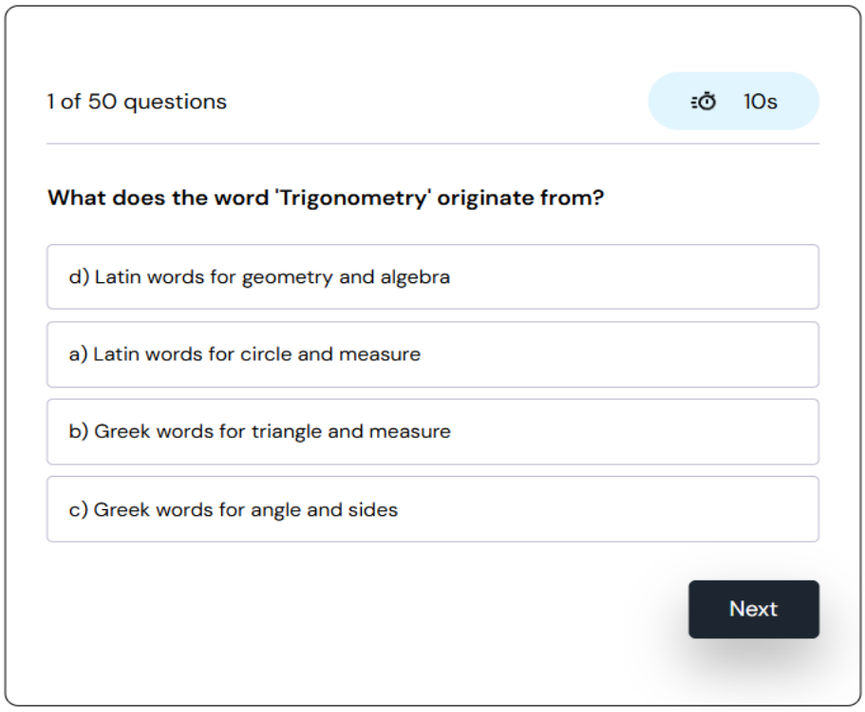


*Please see **Figure 3.1.1:** Overview of Interactive Problem-Solving Exercises (Practice Test) across Science, Mathematics, and Reading modules.*

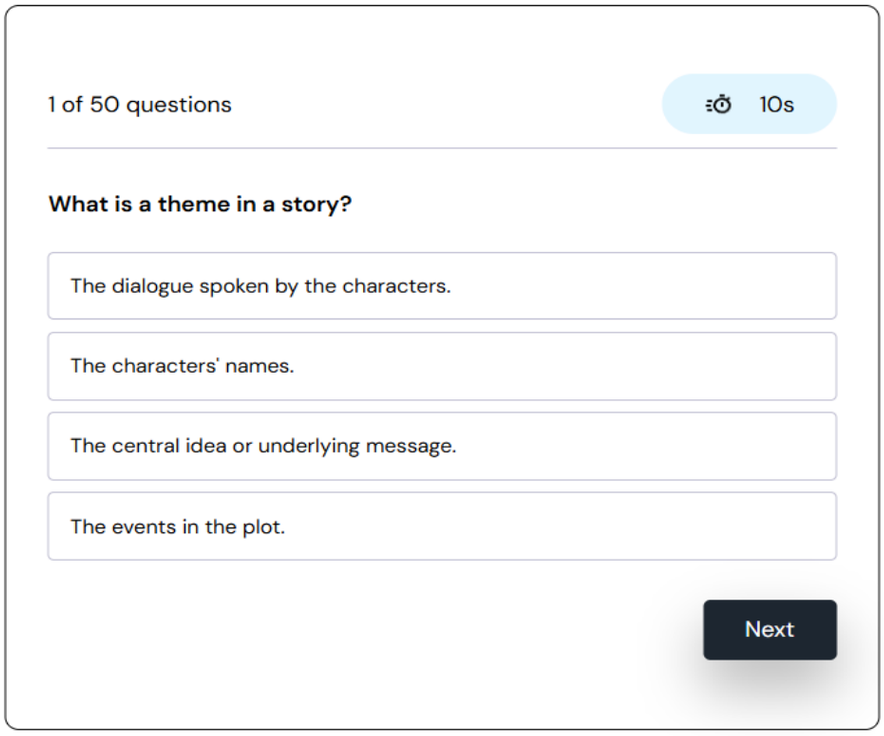
- **Final Tests:**
  The final tests are designed as a comprehensive evaluation tool, assessing students' mastery of the subject matter. These tests determine their readiness to advance to the next level and tackle more complex applications in Mathematics, Science, and Reading.
  - *Visual Representation:*
    Please see Figures 3.1.2a, 3.1.2b, and 3.1.2c for subject-specific views of the Final Tests.
    These images demonstrate the structured and challenging layouts for Mathematics, Science, and Reading modules. They highlight the advanced problem-solving scenarios designed to assess comprehensive understanding and prepare learners for more complex academic tasks.



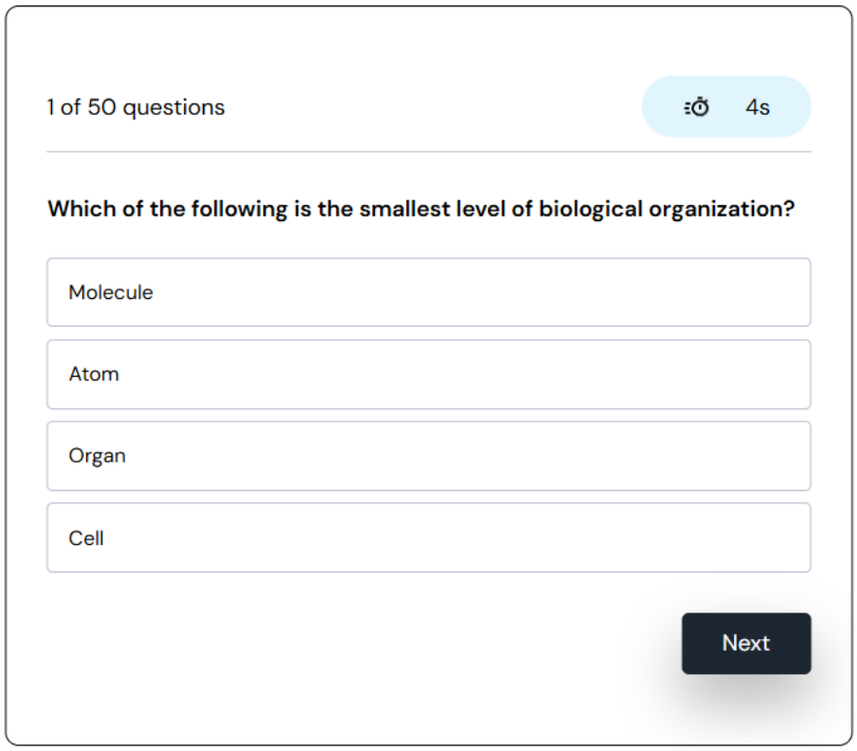
*Please see **Final Tests:** Serve as a comprehensive evaluation of the student's understanding of the subject, determining their readiness for the next level.*



*Please see **Final Tests:** Serve as a comprehensive evaluation of the student's understanding of the subject, determining their readiness for the next level.*



*Please see **Final Tests:** Serve as a comprehensive evaluation of the student's understanding of the subject, determining their readiness for the next level.*



*Please see **Final Tests:** Serve as a comprehensive evaluation of the student's understanding of the subject, determining their readiness for the next level.*

3.2 Engaging Lectures

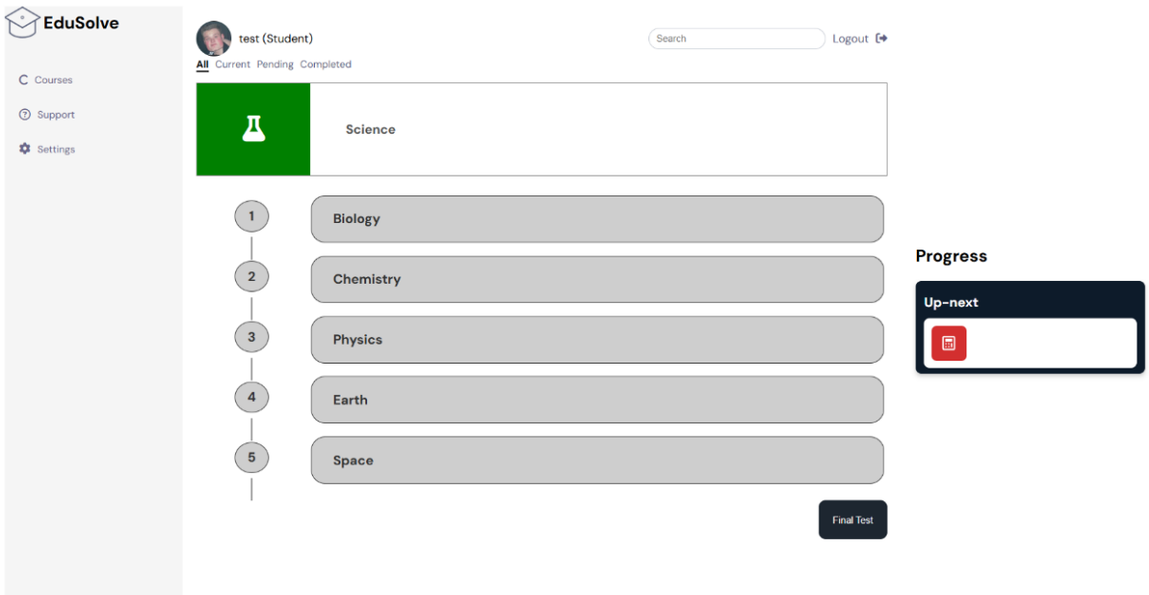
Our engaging lectures offer a comprehensive and interactive learning experience across core subjects: **Science, Mathematics, and Reading**. Through a blend of multimedia resources and dynamic presentations, these lectures are designed to spark curiosity, foster critical thinking, and make learning an enjoyable process.

Key Features:

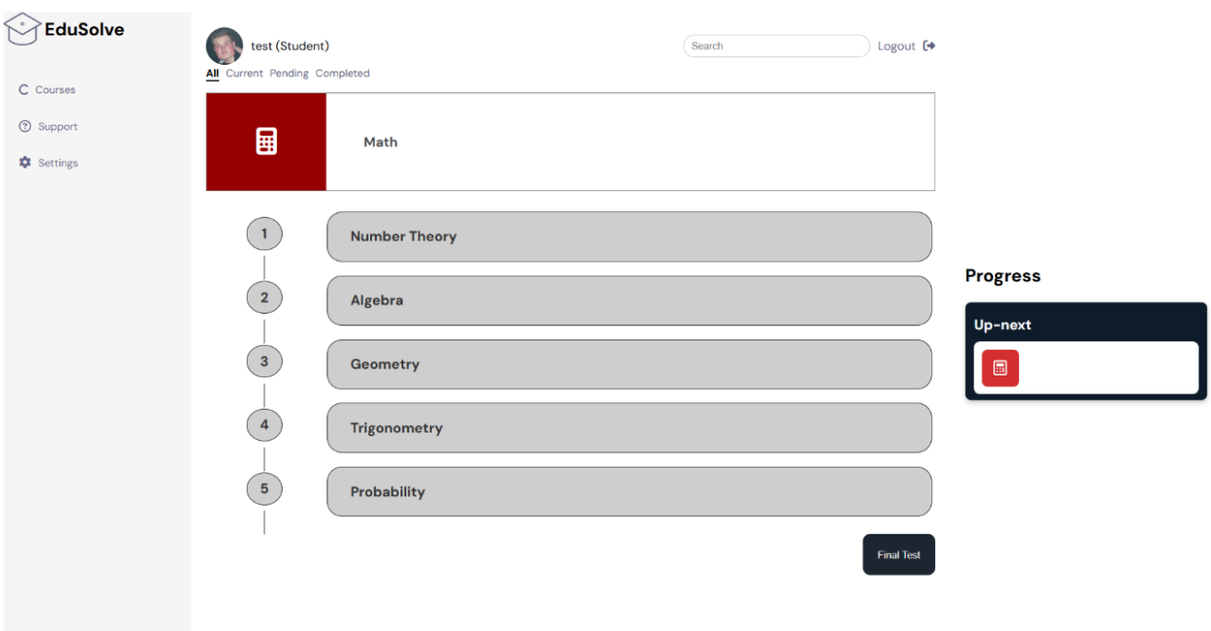
- **Science**: Explore the wonders of the natural world through captivating experiments, virtual field trips, and data-driven analysis.
- **Mathematics**: Master essential mathematical concepts with interactive problem-solving, real-world applications, and engaging visual aids.
- **Reading**: Develop strong language skills through engaging storytelling, creative writing exercises, and interactive discussions.

By combining traditional lecture formats with innovative digital tools, our lectures offer a truly immersive and effective learning experience.

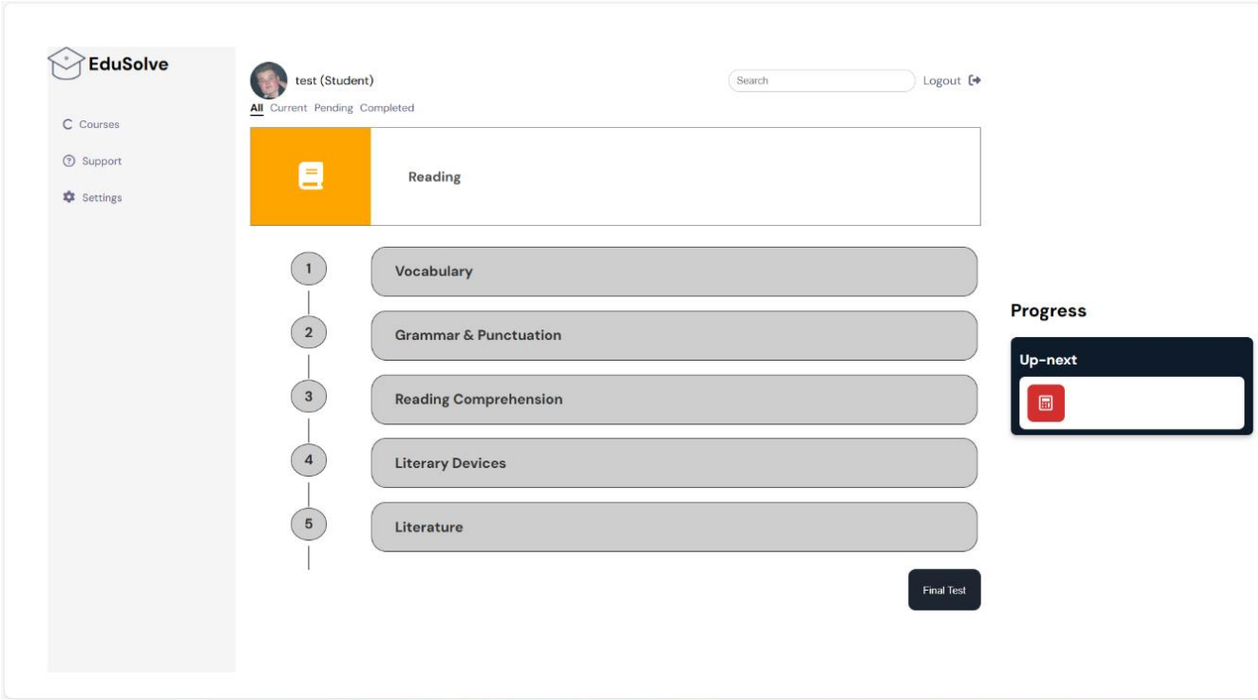
Please see **Figure 3.2.1** for an overview of the engaging lectures, illustrating the structured layout and interactive elements within each subject area.



Please see Figure 3.2.1 for an overview of the Science modules. This image illustrates the structured layout and ease of navigation within the Science subject area.



Please see Figure 3.2.2 for an overview of the Mathematics modules. This image illustrates the structured layout and ease of navigation within the Mathematics subject area.



Please see Figure 3.2.3 for an overview of the Reading modules. This image illustrates the structured layout and ease of navigation within the Reading subject area.

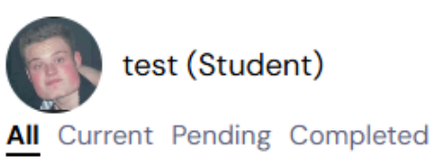
3.3 Progress Tracking

Performance Tracking:

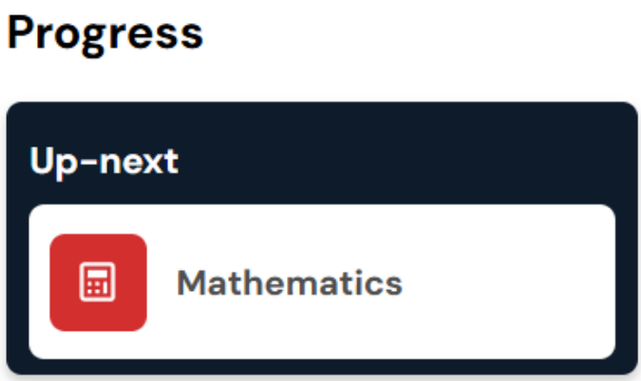
The Edu Solve system continuously tracks student performance, providing a dynamic view of their progress across various subjects. This feature ensures that students are assessed at multiple points in their learning journey, enabling real-time insights into their strengths and areas requiring improvement.

- **Progress Bars:**
  One of the key features of the performance tracking system is the use of **progress bars**. These visual indicators display the current status of a student's progress in a subject or module. The bars are segmented into three distinct states:
  - **Current:** This section of the progress bar shows how far the student has progressed in the current topic or module. It provides a visual representation of the work already completed.
  - **Pending:** This section indicates the remaining content or tasks yet to be completed. It shows students which areas are still pending, giving them a clear roadmap of what's left to cover.
  - **Completed:** Once a student finishes a module or task, the "completed" section fills in, signifying that they have successfully finished that portion of their learning. This provides a sense of accomplishment and motivates students to continue their studies.

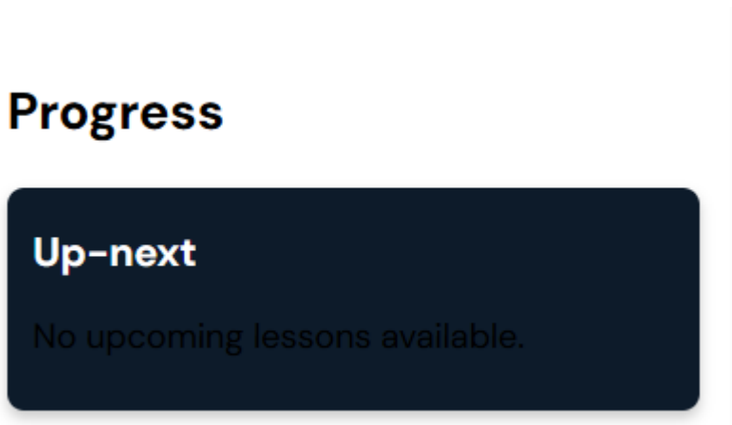
Please see Figure 3.3.1 for a sample view of the progress bar, which illustrates how the current, pending, and completed sections visually represent student progress within the Edu Solve system.



*Please see Figure 3.2.*1 for a sample view of the progress bar



Please see Figure 3.2.1 for a sample view of the progress bar



Please see Figure 3.2.1 for a sample view of the progress bar

3.4. Unlockable Lectures

The **Unlockable Lectures** feature within EDU SOLVE is designed to provide students with an engaging and progressive learning experience. It allows students to access educational content in a structured manner, where certain lessons or lectures become available based on their progress in earlier sections. The system is designed to help students unlock new material as they complete previous tasks, ensuring they build knowledge step-by-step.

Please refer to **Figure 3.4.1** for a sample view of the **Unlockable Lectures**, which illustrates how the current, pending, and completed sections visually represent student progress within the Edu Solve system.

Key Features of Unlockable Lectures:

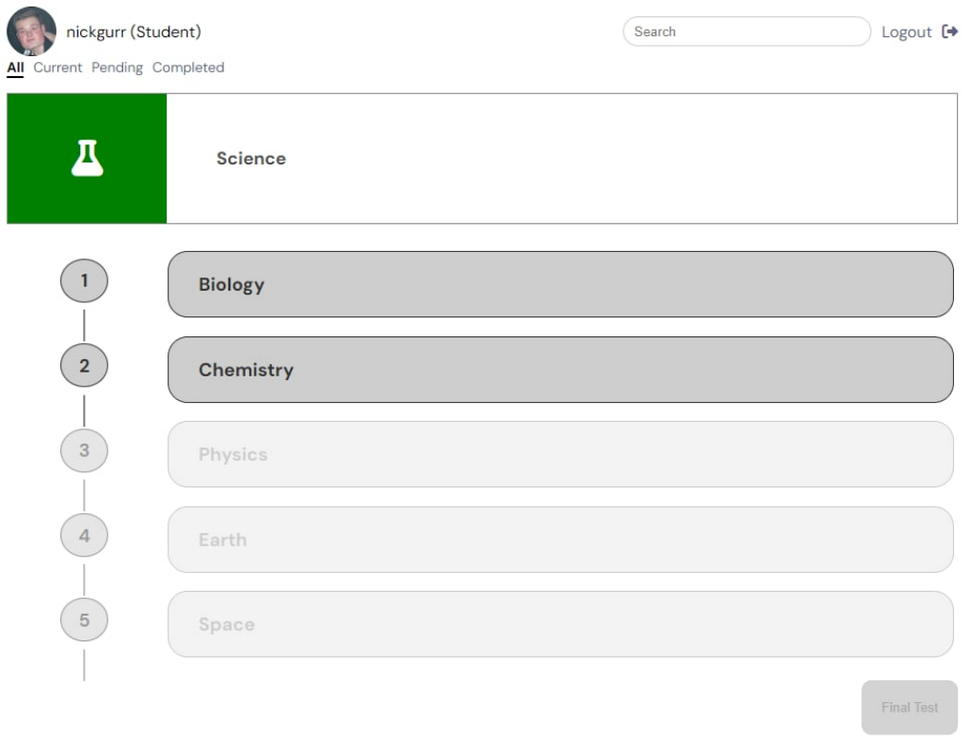
- **Current Lectures:**
  These are the lectures that students are actively working on. They are typically unlocked based on the completion of prior sections or based on predefined milestones within the curriculum. The current section is marked as "in progress" and may include interactive exercises, video content, or reading material.

- **Pending Lectures:**
  Pending lectures are those that will become available once the student successfully completes the current section. These lectures are locked and cannot be accessed until the student progresses through the prerequisite content. Pending lectures are visually indicated in the system with a "locked" status, ensuring students understand that further progress is required.

- **Completed Lectures:**
  Once a student has finished a lecture, it is marked as "completed" in the system. The completed section is highlighted to reflect the student's academic progress. This helps students track their learning journey and see how much content they have already mastered. Completed lectures also serve as a foundation for unlocking the next set of lectures.

  The unlocking mechanism provides a clear structure for students, motivating them to complete tasks before moving on to more advanced content. This approach not only ensures students grasp foundational concepts before advancing but also encourages continuous learning and achievement.

  By structuring content in this manner, EDU SOLVE helps students progress at their own pace, keeping them engaged, challenged, and motivated throughout their learning journey.



*Please see Figure 3.4.*1 for a sample of the unlockable lectures

3.6 Account Management

The Account Management feature enhances the system's usability and security by allowing users to update their account credentials directly within the platform. This feature ensures that users can manage their personal information securely and efficiently.

Key Features:

- **Update Username:** Users can change their username to reflect their preferences or correct errors during registration.
- **Update Password:** Users can enhance security by updating their passwords periodically.

Workflow:

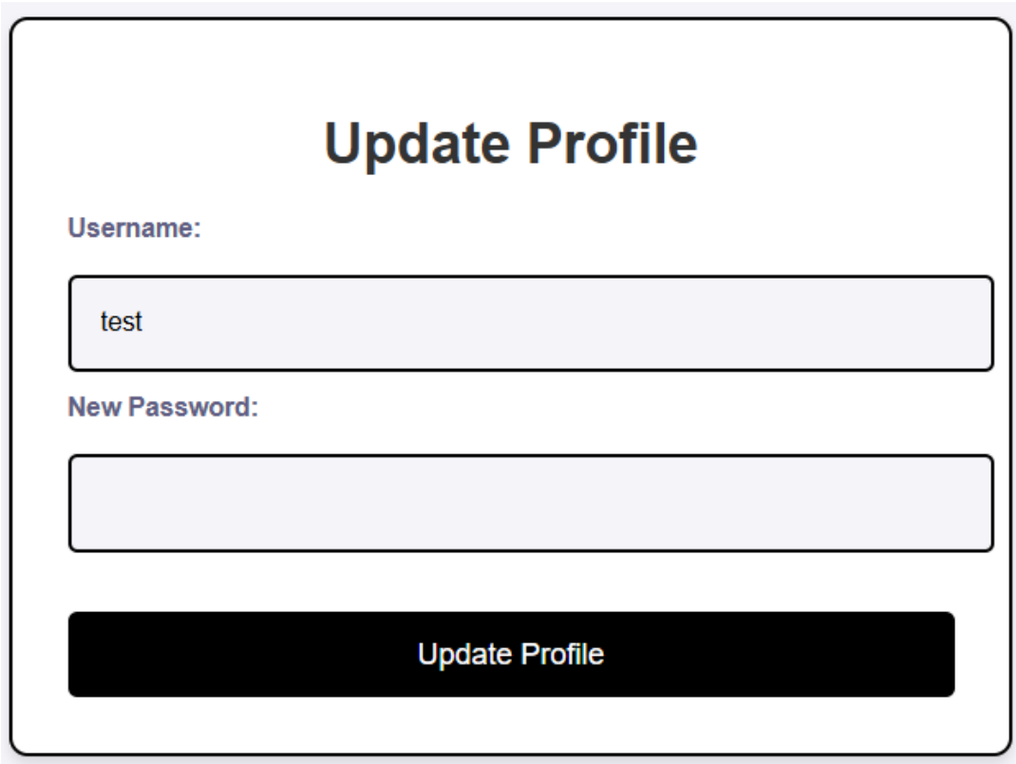
1. Navigate to the **Settings** menu.
2. Select either **Update Username** or **Update Password**.
3. Input the required information in the respective fields.
    - For username updates: Enter the new username.
    - For password updates: Enter the current password and the new password (with confirmation).
4. Submit the changes. The system validates the inputs and applies the updates upon successful verification.

Benefits:

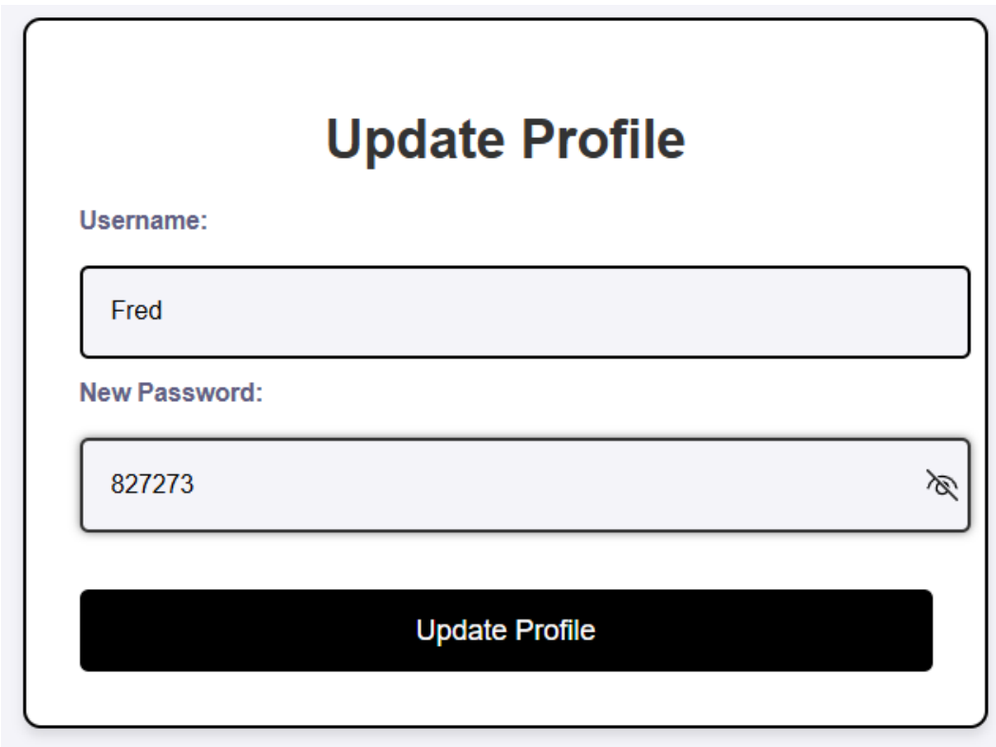
- **Improved Security:** Encourages users to keep their credentials secure and updated.
- **Ease of Use:** A simple and intuitive process ensures all users, regardless of technical ability, can manage their accounts effectively.
- **Enhanced User Control:** Gives users the ability to personalize and secure their accounts.

Figure Reference:

Please see **Figure 3.6.1** and **Figure 3.6.2** for an example of the Settings interface, showcasing the options for updating username and password.



*Please see Figure 3.6.*1 for a sample of the Settings interface



*Please see Figure 3.6.*2 for a sample of the update Settings interface

CHAPTER 4

SYSTEM ARCHITECTURES

4.1 System Workflow

The system workflow for **Edu Solve** begins with user interaction on the front end. Users access the website to navigate through educational content, perform actions like logging in, submitting forms, or viewing resources. These interactions trigger processes in the back end, which handles data processing, authentication, and communication with the database. The workflow ensures a seamless exchange of information between the user interface, application logic, and database storage.

4.2 Front-End Architecture

The front end of Edu Solve is designed with a focus on providing a seamless and engaging user experience. The architecture prioritizes responsiveness, interactivity, and ease of navigation. It uses a combination of core technologies and features to ensure that users have a smooth and effective learning experience.

Technologies Used

1. **HTML (HyperText Markup Language)**
    - Serves as the backbone of the website, structuring the content and elements.
    - Key Implementation:
        - Used for creating static elements like login forms, subject lists, and quiz pages.
        - Example: Structuring input fields for user registration, quiz questions, and content display areas.
2. **CSS (Cascading Style Sheets)**
    - Enhances the visual appeal of the platform while ensuring responsiveness across devices.
    - Key Implementation:
        - Designed a consistent color scheme and typography aligned with the educational theme.
        - Applied media queries for responsive design, adapting layouts for mobile, tablet, and desktop screens.

        - Example: The dashboard grid layout adjusts based on screen size—displaying 3 columns on desktops and stacking them vertically on smaller devices.

3. **JavaScript**
    - Adds dynamic behavior and interactivity to the website, making the learning experience more engaging.
    - Key Implementation:
        - Form Validation: Ensures user input is correct, such as email format validation on registration forms.
        - Interactive Features: Powers dynamic quizzes, dropdown menus, and real-time progress tracking.
        - Example: Real-time quiz result updates and filtering subjects based on user selection.

Core Features of the Front End

1. **Responsive User Interface**

- Optimized for laptops and desktop computers.
- Example: The navigation menu is designed for larger screens, ensuring a clear and accessible layout for users on desktop and laptop devices.
- Please see Figure 4.1 for the design details.

2. **Interactive Components**
    - Implements engaging features like:
        - Progress Bars: Show users' progress during quizzes.
        - Pop-ups and Modals: For notifications, password resets, or providing hints during quizzes.
        - Example: A modal appears after completing a quiz, showing performance results.

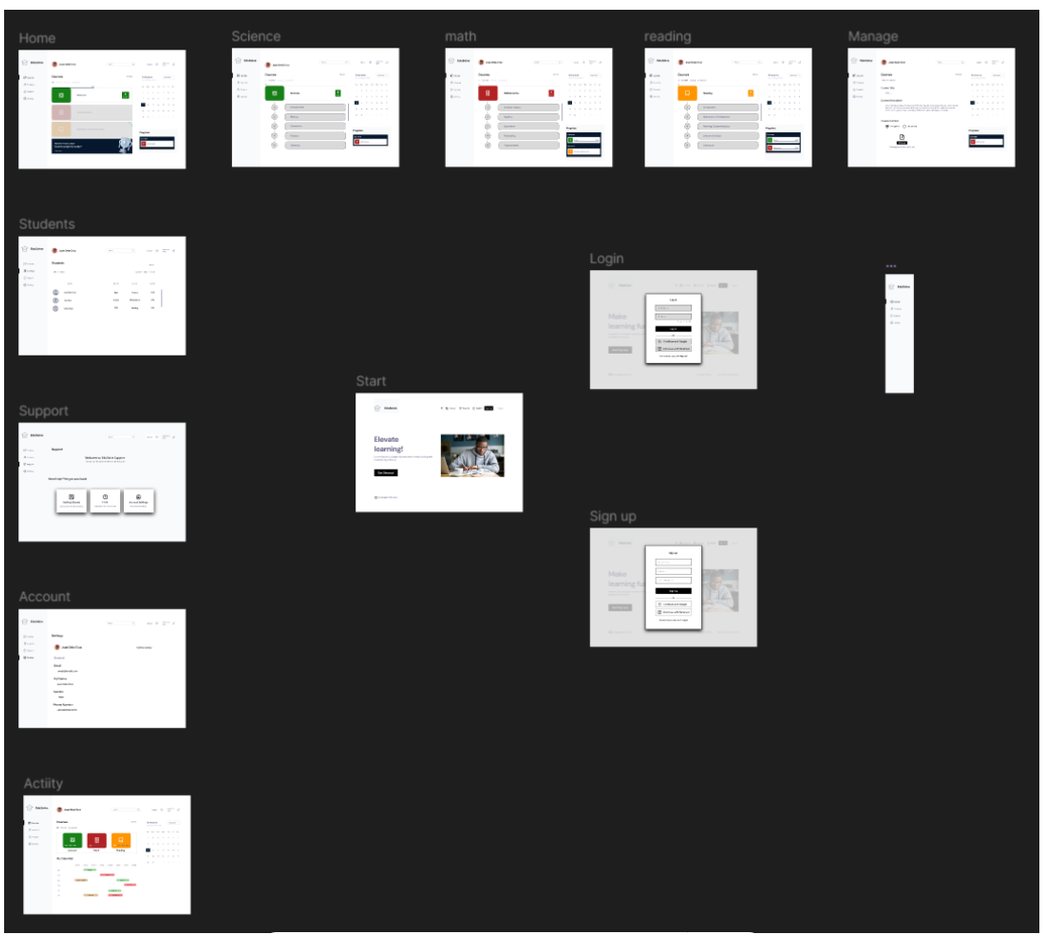
3. **Seamless Communication with the Back End**

   - The front end communicates with the back end using AJAX, ensuring smooth real-time updates without page reloads.
   - Example: When users unlock a new subject, the front end sends a request to the back end, and the updated list of subjects is displayed dynamically

4. **Prototyping and Design**

   - Figma was used for prototyping and refining the user interface before actual implementation.
     - Example: The dashboard layout, quiz screens, and onboarding flow were tested and iterated in Figma to ensure usability and consistency**.**

Please see Figure 4.1 for the initial wireframe design the following figures for the wireframes and prototypes that demonstrate the user interface and flow:



Please see Figure 4.1 for the initial wireframe design. This image illustrates the layout and flow of the front-end interface, showcasing how users interact with the system from login to dashboard.

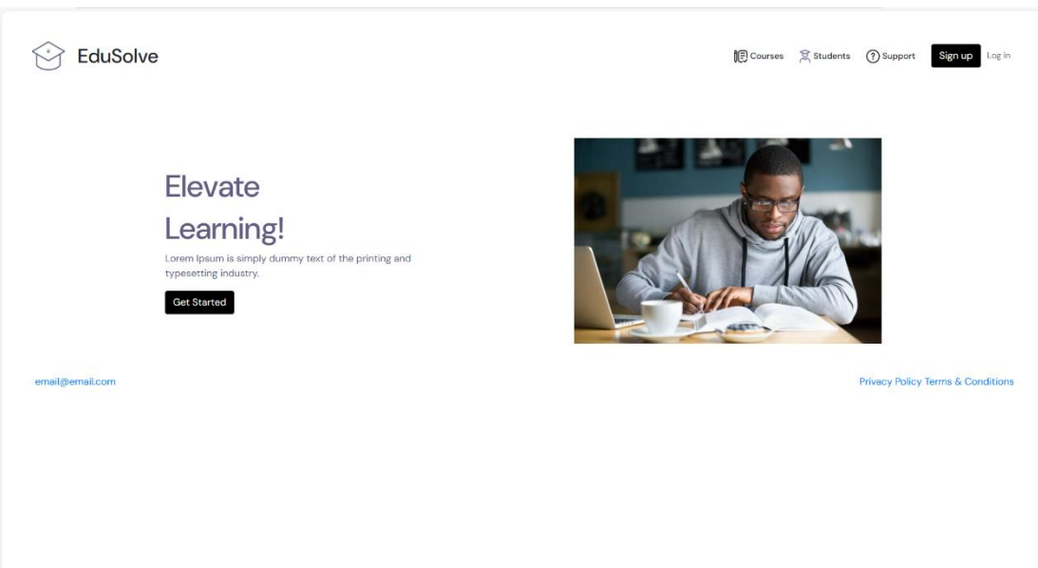
User-Centric Design Principles

The front-end architecture follows a user-centric approach, ensuring the following:

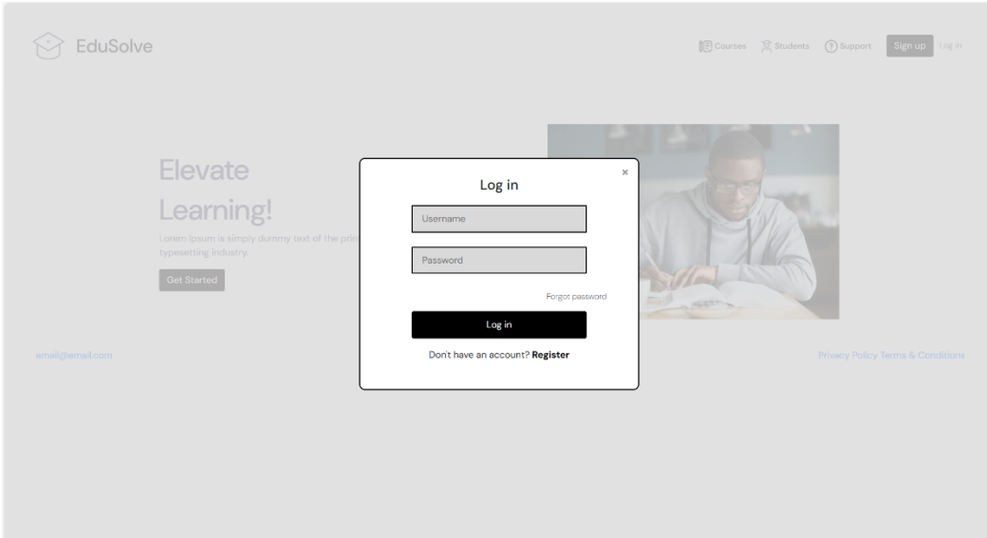
- **Intuitive Navigation:** Users can quickly access their assigned subjects and switch between dashboards, quizzes, and progress trackers.
- **Seamless Flow:** Clear, responsive transitions between pages (e.g., Login → Dashboard → Progress Tracking) guide users through the app.
- **Visual Feedback:** Provides clear feedback to users, such as quiz completion bars or form error highlights.
    - Example: When a quiz is completed, users see a completion bar and score feedback.

Visuals:

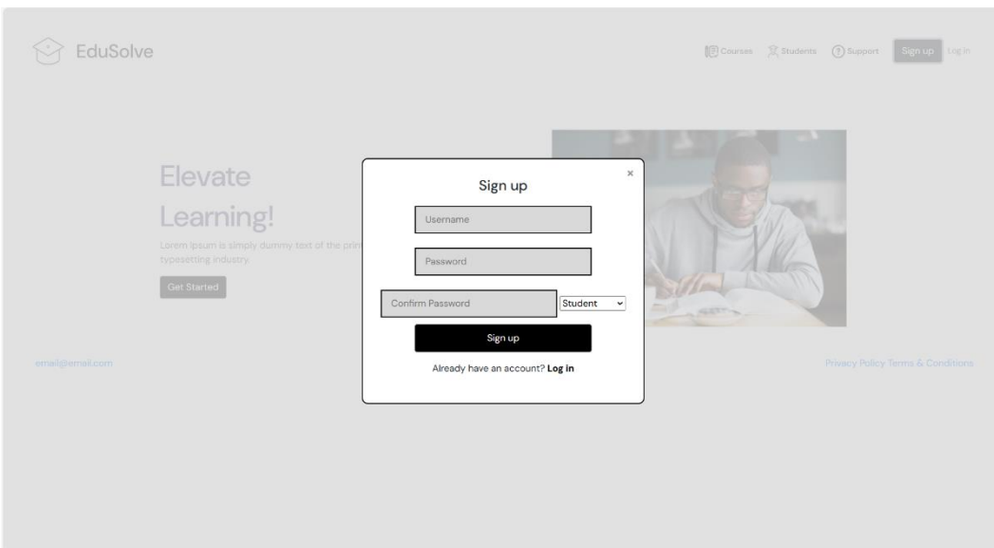
Please see the following figures for the wireframes and prototypes that demonstrate the user interface and flow:



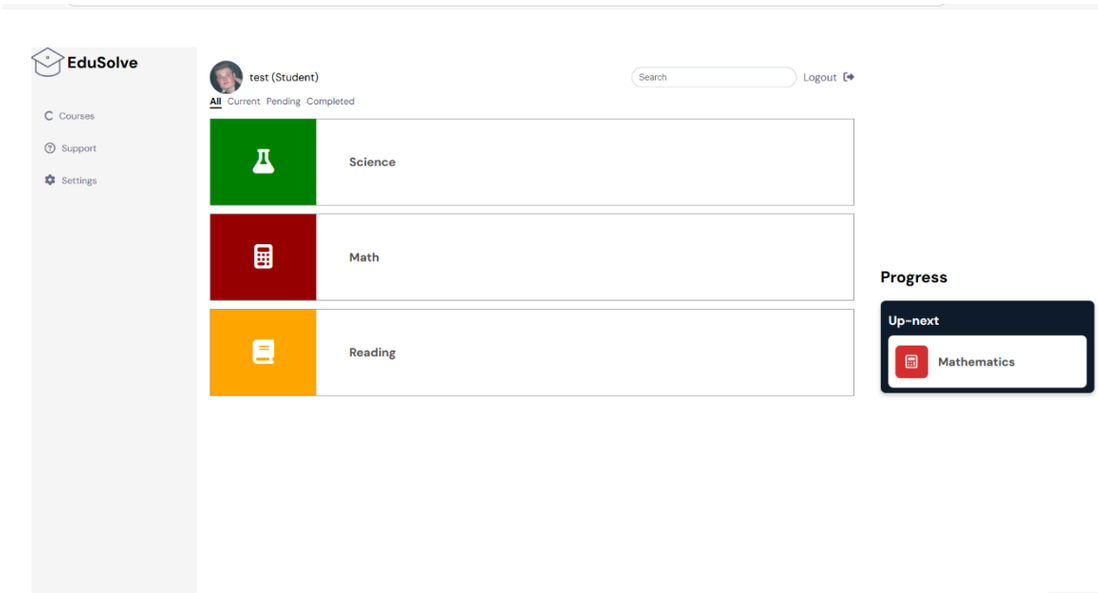
- Figure 4.1: *Homepage - The entry point of the system, where users can initiate their journey and log in or sign up*.



- Figure 4.2: *Login Page - Where users input their credentials to access the system*.



- Figure 4.3: *Sign Up Page - New users can register for the system here*.



- Figure 4.4: *Dashboard - The main interface users see after logging in, where they can access their subjects, progress, and other features*.

These images showcase how the interface elements are arranged and how users interact with them.

4.3 Back-End Architecture

The back end of Edu Solve is implemented using PHP, which processes server-side logic and ensures secure communication with the database. The following are the key responsibilities of the back end:

Core Functionalities

- **User Authentication**
  - User Login: Secure login system to authenticate users.
  - User Registration: Allow new users to create accounts.
  - Password Reset: Feature to recover account access securely.
- **Content and Quiz Management**
  - Upload lessons, quizzes, and exercises for students.
  - Track quiz results and provide progress reports for each user.
- **Student Management**
  - Add, update, delete, and view student profiles.
  - Assign subjects to students and manage their access based on their progress.
- **Portfolio Management**
  - Allow students to upload completed assignments and projects.
  - Manage descriptions for projects and integrate portfolios with external platforms.
- **Request and Communication Management**
  - Students can submit, update, view, and delete learning requests.
  - In-app messaging system for communication between users and administrators.
  - File sharing for submitting assignments or providing references**.**

Additional Features

- Learning Path and Achievement Tracking
  - Monitor students' progress through milestones, quizzes, and subject unlocking.
- **Security and Efficiency**
  - Ensure data integrity through secure communication with the database and validation of inputs.
  - Optimize server-side logic for efficient handling of multiple requests.

4.4 Database Design

The Edu Solve database is built using MySQL and is structured to store and manage user data, educational content, and progress tracking.

4.4.1 Key Components of the Database

- **Users Table:** Stores user credentials and profile details.
- **Subjects Table:** Contains information about all available subjects.
- **Student_Subjects Table:**
  - Tracks the relationship between students and their assigned subjects.
  - The is_unlocked column indicates whether a student has access to a specific subject.
- **Quizzes Table:** Stores quiz questions, answers, and results for progress tracking.
- **Requests Table:** Maintains student-submitted requests for additional materials or assistance.

4.4.2 Relationships Between Tables

The database is designed with foreign key relationships to ensure data consistency. For example, the user_id column in the student_subjects table links to the Users table, connecting students to their assigned subjects. These relationships help manage data integrity and facilitate efficient queries.

4.4.3 Entity-Relationship Diagram (ERD)

Please refer to Figure 4.4.1 for the Entity-Relationship Diagram (ERD). This diagram visually represents the relationships between the core tables in the Edu Solve database and illustrates how data is interconnected across the system.

Detailed Explanation of the ERD:

- **Users Table:**
  The central table in the database, the Users table, stores all user-related data, including their credentials and profile details. The user_id serves as the primary key, uniquely identifying each user.

- **Subjects Table:**
  The Subjects table contains data about the available subjects (e.g., Science, Mathematics, English). Each subject is assigned a unique subject_id as its primary key. The subject_name and description fields provide more information about the subject.

- **Student_Subjects Table:**
  This table acts as a junction table to track the relationship between users (students) and the subjects assigned to them. Each entry links a user (via student_id) to a subject (via subject_id).
  The student_subject_id is the primary key of this table.
  Additionally, the is_unlocked column helps track whether a student has access to a particular subject, and the completion_status column can indicate progress.

  - Foreign Keys:

    - student_id is a foreign key that references the Users table (user_id).

    - subject_id is a foreign key that references the Subjects table (subject_id).

- **Quizzes Table:**
  The Quizzes table stores quiz-related data. Each quiz is identified by a unique quiz_id, and it contains fields for the question text, available answer choices, and the correct answer.

- **Quiz_Results Table:**
  This table tracks each quiz attempt made by users. Each entry records the user's score on a particular quiz.
  The result_id is the primary key of this table, while user_id and quiz_id serve as foreign keys**.**

  - user_id references the Users table, linking the result to a specific user.

- quiz_id references the Quizzes table, indicating which quiz the result corresponds to.
- The score field records the user's performance on the quiz, and the timestamp tracks when the quiz was taken**.**

- **Requests Table:**
  This table stores requests submitted by students for additional learning materials or assistance. The request_id is the primary key.
  - student_id is a foreign key linking to the Users table, connecting the request to a specific student.
  - The request_type field categorizes the request (e.g., additional materials or specific queries), and the status field tracks the progress of the request (e.g., pending, resolved).

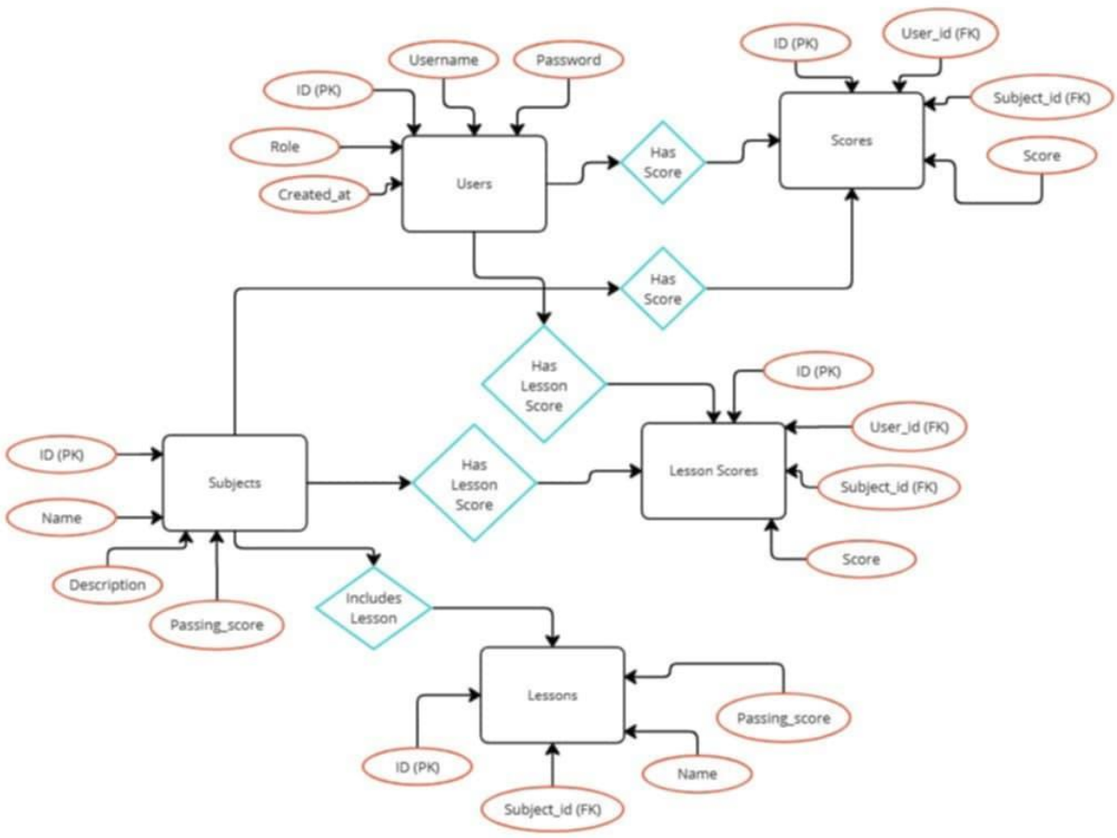


Figure 4.4.1: Entity-Relationship Diagram (ERD) for Edu Solve Database

4.4.4 Schema (Table Structure)

The schema defines how each table is structured, with columns and relationships (foreign keys) between them. Below is the schema definition based on the ERD:

Users Table

```
CREATE TABLE Users (
    user_id INT AUTO_INCREMENT PRIMARY KEY,
    username VARCHAR(255) NOT NULL,
    password_hash VARCHAR(255) NOT NULL,
    email VARCHAR(255) NOT NULL,
    first_name VARCHAR(255),
    last_name VARCHAR(255),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Subjects Table:

```
CREATE TABLE Subjects (
    subject_id INT AUTO_INCREMENT PRIMARY KEY,
    subject_name VARCHAR(255) NOT NULL,
    description TEXT
);
```

Student_Subjects Table:

```
CREATE TABLE Student_Subjects (
    student_subject_id INT AUTO_INCREMENT PRIMARY KEY,
    student_id INT,
    subject_id INT,
    is_unlocked BOOLEAN DEFAULT FALSE,
    completion_status VARCHAR(50),
```

```
    FOREIGN KEY (student_id) REFERENCES Users(user_id),
    FOREIGN KEY (subject_id) REFERENCES Subjects(subject_id)
);
```

Quizzes Table:

```
CREATE TABLE Quizzes (
    quiz_id INT AUTO_INCREMENT PRIMARY KEY,
    question TEXT NOT NULL,
    answer_choices TEXT NOT NULL,
    correct_answer VARCHAR(255) NOT NULL
);
```

Quiz_Results Table:

```
CREATE TABLE Quiz_Results (
    result_id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT,
    quiz_id INT,
    score INT,
    timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (user_id) REFERENCES Users(user_id),
    FOREIGN KEY (quiz_id) REFERENCES Quizzes(quiz_id)
);
```

Requests Table:

```
CREATE TABLE Requests (
    request_id INT AUTO_INCREMENT PRIMARY KEY,
    student_id INT,
    request_type VARCHAR(255),
    status VARCHAR(50),
    FOREIGN KEY (student_id) REFERENCES Users(user_id)
);
```

4.4. 5 Visual Representation in the ERD:

The ERD diagram visually shows how each table connects with others using foreign keys. The relationships between tables are represented with lines connecting them, with annotations indicating the nature of the relationship (e.g., one-to-many). Foreign key relationships are denoted by arrows or lines pointing to the relevant table, ensuring data integrity and easy navigation through the database structure.

Please refer to Figure 4.4.6 for the Table Schema for the Edu Solve Database This diagram visually represents the relationships between the core tables in the Edu Solve database and illustrates how data is interconnected across the system.

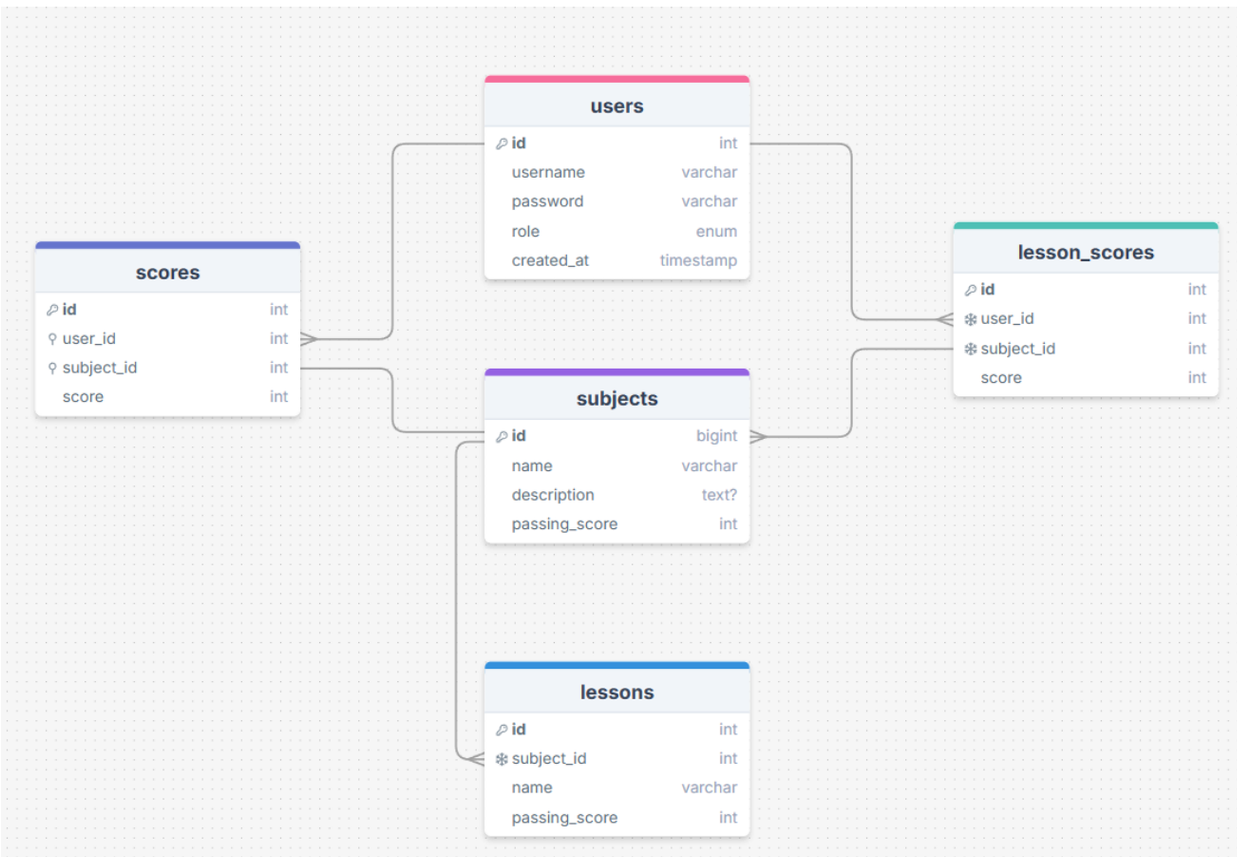


Figure 4.4.7: Table Schema for the Edu Solve Database

4.4.6 Query Optimization

- Queries are optimized to ensure fast access to user progress, educational materials, and quiz data.
- Indexes, Indexes are implemented on frequently queried columns, such as user_id and subject_id, to enhance database performance.

4.4. 7 Relationships Between Tables:

- **Users ↔ Student_Subjects:**
  The relationship between the Users and Student_Subjects tables is one-to-many. A single user (student) can be assigned to multiple subjects, but each entry in the Student_Subjects table links to only one user. The student_id field in Student_Subjects references the Users table.
- **Subjects ↔ Student_Subjects:**
  The relationship between the Subjects and Student_Subjects tables is also one-to-many. A subject can be assigned to multiple students, but each record in the Student_Subjects table relates to only one subject. The subject_id field in Student_Subjects references the Subjects table.
- **Users ↔ Quiz_Results:**
  The relationship between the Users and Quiz_Results tables is one-to-many. A user can have multiple quiz results, as they may attempt several quizzes, but each result in the Quiz_Results table refers to a specific user.
- **Quizzes ↔ Quiz_Results:**
  The relationship between the Quizzes and Quiz_Results tables is one-to-many. A quiz can have multiple results, with each quiz attempt by different users being recorded in Quiz_Results.
- **Users ↔ Requests:**
  The relationship between the Users and Requests tables is one-to-many. A user (student) can submit multiple requests, but each request is linked to a single user.

In summary, this ERD helps you understand how user data, subject assignments, quiz results, and requests are all linked together in the database, enabling efficient data retrieval and management

CHAPTER 5

METHODOLOGY

5. Methodology

This section outlines the methodology followed during the development of the Edu Solve system. The process was divided into distinct phases to ensure systematic planning, development, and testing.

5.1. Development Approach

The Agile Development Model was employed for this project. This approach allowed for iterative design, development, and testing cycles, ensuring flexibility and responsiveness to feedback.

5.2. Phases of Development

1. **Requirement Analysis:**
   - Identified core features such as quizzes, progress tracking, and user management.
   - Conducted brainstorming sessions with stakeholders to gather input for features and design.
2. **Design:**
   - Created initial wireframes and mockups using Figma to design the system's user interface.
   - Developed an Entity-Relationship Diagram (ERD) to structure the database design.
   - Designed a flowchart to outline user navigation and system behavior.
3. **Implementation:**
   - Frontend developed using HTML, CSS, and JavaScript.
   - Backend functionality implemented using PHP for server-side logic and MySQL for database management.
   - Integrated user authentication, quiz management, and progress tracking features.
4. **Testing and Debugging:**
   - Conducted unit testing for individual modules (e.g., quizzes, user authentication).
   - Performed system testing to ensure the system worked as a cohesive unit.
   - Addressed bugs and optimized the system for performance.
5. **Deployment:**

- Deployed the system on a local or online server for demonstration and usability testing.

6. **Evaluation:**
    - Collected user feedback during testing to refine features and improve usability.

5.3. Tools and Technologies Used

- Design: Figma for wireframes and prototype development.
- Development: PHP, MySQL, HTML, CSS, JavaScript.
- Version Control: GitHub for version control and collaboration.
- Testing: Manual testing and debugging tools for performance optimization.

5.4. Iterative Process

Throughout the development, an iterative process was followed, revisiting and refining the system based on stakeholder feedback and test results. This approach ensured that the system met user needs and performed reliably.

CHAPTER 6

DEPLOYMENT STEPS

- The successful deployment of EDU SOLVE involves a series of well-coordinated steps that ensure the platform is accessible, functional, and secure for its users. The following outlines the key deployment steps taken to bring EDU SOLVE from development to production:

6.1 User Guide

- The User Guide serves as a comprehensive resource to assist both administrators and students in using EDU SOLVE effectively. It provides clear instructions on how to navigate the platform, access its features, and make the most of its educational tools. Below are the key sections:

- **Account Setup and Registration**:
  This section explains how users—students and administrators—can create an account and access the platform. It provides instructions on registering, verifying email addresses, and setting up secure passwords.

- **Navigating the Platform**:

- **Dashboard Overview**: Description of the main dashboard and its components, including available subjects, progress tracking tools, and quiz options.

- **Subject Access**: Steps for students to select and access different subjects, engage with exercises, and view lectures.

- **Progress Tracking**: Instructions for monitoring academic performance, viewing quiz results, and assessing progress.

- **Progress Monitoring**:
  A critical part of EDU SOLVE is enabling students to track their academic performance. This section includes:

- Viewing quiz scores and completion status for each subject.

- Understanding progress indicators like "unlocked" subjects and completed exercises.

- Accessing historical data for improvement tracking.

6.2 Web Hosting Setup

- To ensure EDU SOLVE is reliably accessible, the platform has been deployed on a secure and scalable web hosting service. Key steps include:

- **Server Configuration**: Setting up the environment on a cloud-based platform to host the web application.
- **Database Setup**: Installing and configuring MySQL for data management.
- **Security Measures**: Implementing SSL certificates, firewalls, and encryption to protect user data.

6.3 Platform Testing and Quality Assurance

- Before launch, the platform underwent extensive testing to meet performance and usability standards:
- **Functional Testing**: Verifying features like registration, login, subject access, quizzes, and progress tracking.
- **Load Testing**: Simulating high traffic to test performance under heavy loads.
- **Bug Fixes and Optimization**: Addressing issues and improving platform speed and reliability.

6.4 Launch

- Upon successful testing, EDU SOLVE was officially launched. Key actions include:
- **Notification to Stakeholders**: Informing stakeholders, including teachers, students, and administrators, about the platform's availability.
- **Public Access**: Enabling student enrollment and platform engagement.

6.5 Ongoing Support and Maintenance

- Post-launch efforts include:
- **User Support**: Providing troubleshooting and technical assistance.
- **Software Updates**: Regular updates to fix bugs, improve functionality, and add features.
- **System Monitoring**: Ensuring optimal uptime, security, and responsiveness.
- based on user feedback.
- System Monitoring: Continuously monitoring the platform's performance to ensure optimal uptime, security, and responsiveness.

CHAPTER 7

CONCLUSION

Conclusion

EDU SOLVE represents a significant step forward in addressing the educational challenges faced by Grade 10 students in the Philippines. By leveraging technology, this web-based platform offers a dynamic and personalized learning experience that aims to improve students' proficiency in critical subjects such as Mathematics, Science, and Reading. Through interactive problem-solving exercises, engaging lectures, and real-time progress tracking, EDU SOLVE empowers students to take charge of their learning journey and overcome barriers to academic success.

The platform’s design incorporates user-centric features, such as account management and progress tracking, ensuring that each student’s learning experience is tailored to their individual needs. Furthermore, the system’s integration of modern educational techniques and its ability to adapt to the evolving needs of learners makes it a valuable tool in the pursuit of quality education.

EDU SOLVE is not only a response to the immediate educational gaps highlighted by the 2022 PISA results but also a contribution to the broader mission of improving the quality of education in the Philippines. As it continues to evolve, EDU SOLVE aspires to play a key role in shaping a generation of learners who are prepared to excel both locally and globally. Ultimately, this platform represents a future where technology and education converge to create a more inclusive, effective, and empowering learning environment.

Future Enhancements

- Integration of AI for personalized learning experiences.
- Expansion to other grade levels and subjects.
- Development of a mobile application for offline accessibility.
- Incorporating a collaborative space for peer-to-peer learning.

This documentation outlines the vision, purpose, and functionalities of the EDU SOLVE project while providing a roadmap for future development. Let me know if you need to add or refine any sections further!